SQL-BASED E-COMMERCE DATA ANALYSIS USING PYTHON AND SQLITE

Project Report

Submitted By

Thamindu Kavinda

Data Analytics Intern DecodeLabs

Submitted To

DecodeLabs

Submission Date

2026/05/28

1. Introduction

This project mainly revolves around SQL-based data analysis on an E-commerce Sales dataset using Python, Pandas, and SQLite in Google Colab.

The main objective of this project was to clean the dataset, store it in a database, and perform several SQL operations on it to extract valuable insights. SQL is one of the most popular tools in the field of data analytics for retrieving, selecting, grouping, and analyzing large amounts of data.

Some of the SQL operations performed in this project include SELECT, WHERE, GROUP BY, ORDER BY, COUNT, SUM, and AVG.

2. Objectives

- To load and inspect the E-Commerce dataset
- To identify and handle missing values
- To check and remove duplicate records
- To create a SQLite database from the cleaned dataset
- To perform SQL-based data analysis
- To generate business insights using SQL queries
- To improve practical knowledge of database analytics

3. Technologies Used

- Python - Data processing
- Pandas - Data manipulation
- SQLite - Database management
- SQL - Data querying and analysis
- Google Colab - Development environment
- Microsoft Excel - Data storage

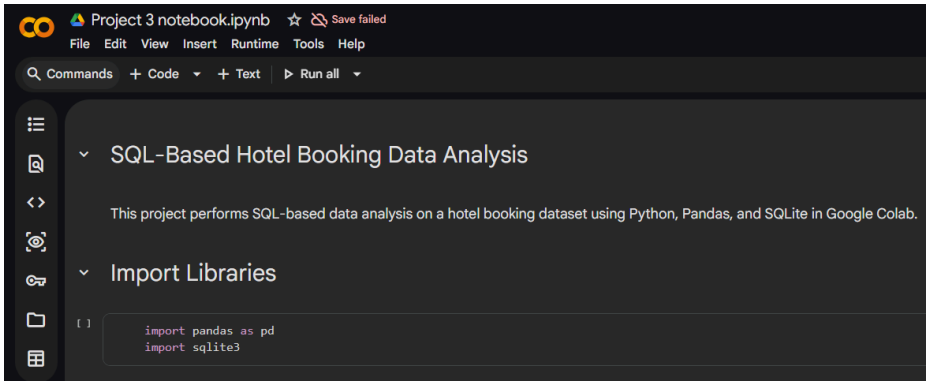
4. Dataset Description

The dataset used in this project contains E-Commerce transaction records. It includes information related to products, customer purchases, payment methods, order status, referral sources, coupon usage, and transaction values.

The dataset was first cleaned and preprocessed before being imported into a SQLite database for SQL-based analysis. The cleaned dataset was then used to perform various analytical queries and generate business insights.

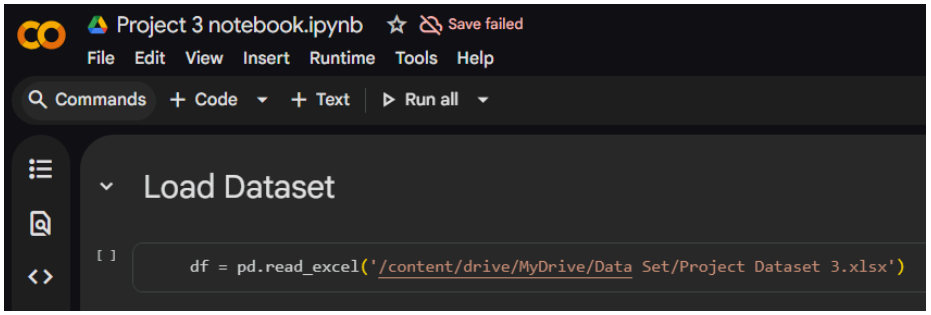
5. Data Cleaning Process

Importing Libraries



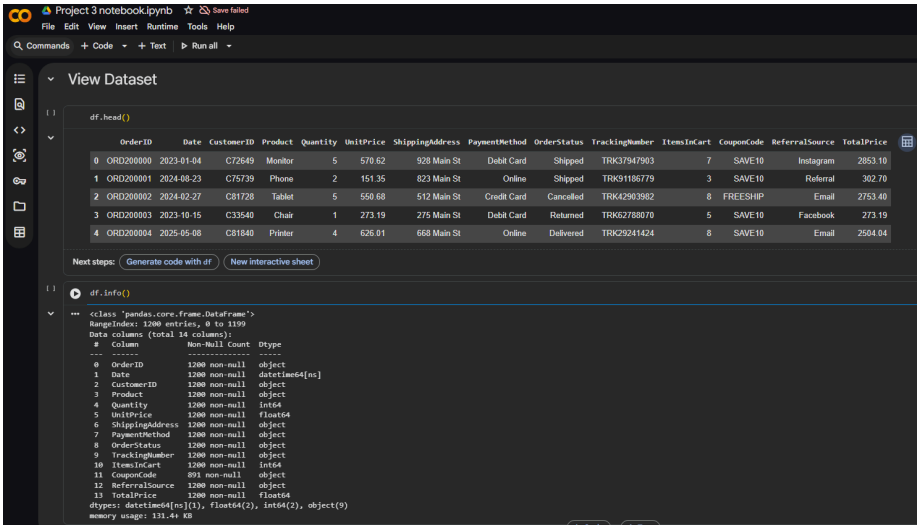
The required Python libraries were imported to perform data cleaning and database operations. Pandas was used for data manipulation, while SQLite was used for database creation and SQL query execution.

Loading the Dataset



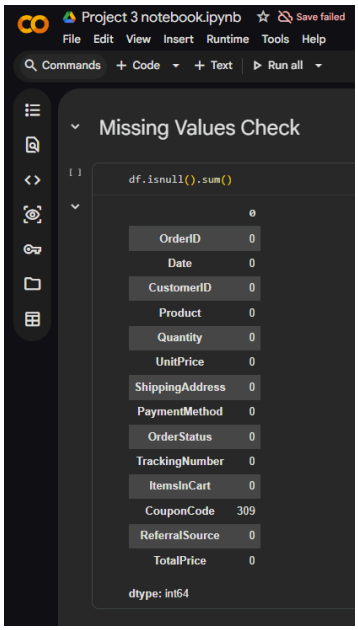
The E-Commerce dataset was loaded into Google Colab using Pandas. This allowed the dataset to be inspected and prepared for further processing.

Dataset Inspection



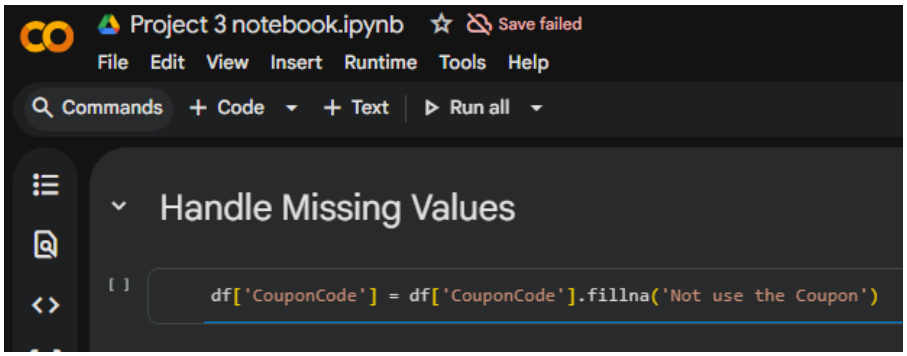
The first few records of the dataset were displayed to understand its structure, column names, and data contents.

Missing Value Detection



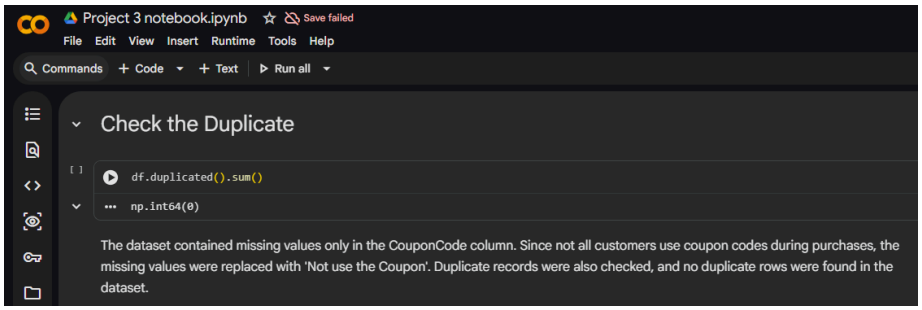
The dataset was examined for missing values to identify incomplete records that could affect the accuracy of the analysis.

Handling Missing Values



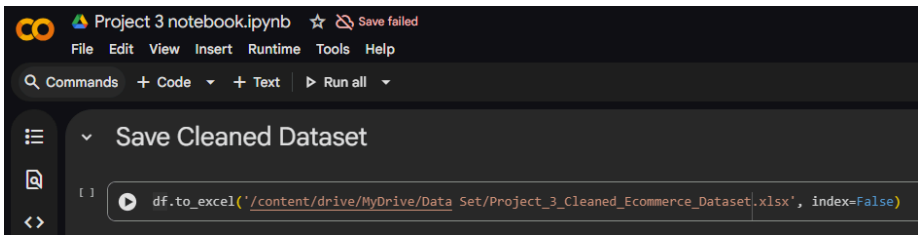
Missing values in the CouponCode column were replaced with an appropriate value to maintain data consistency and improve dataset quality.

Duplicate Record Checking



The dataset was checked for duplicate records to ensure that repeated entries would not affect the accuracy of the analytical results.

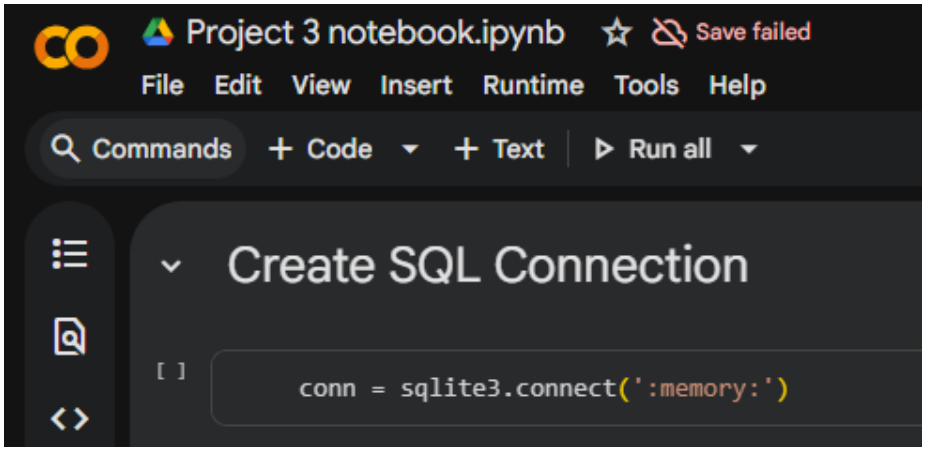
Exporting the Cleaned Dataset



After completing the cleaning process, the dataset was exported as a new file for future analysis and database integration.

6. Database Creation and SQL Integration

Creating SQLite Connection



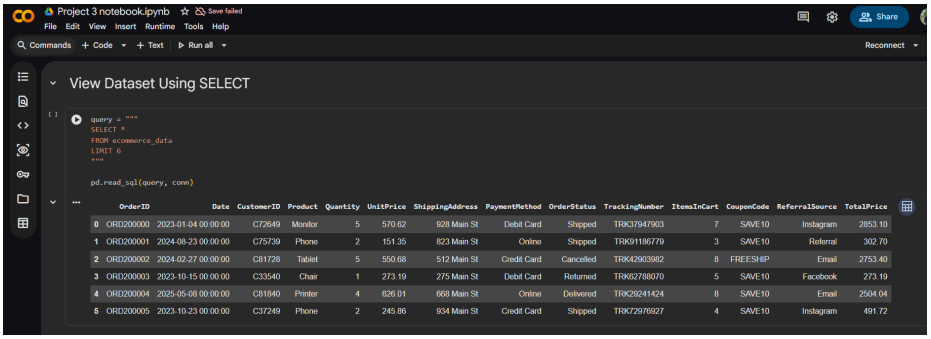
An SQLite database connection was created within Google Colab. This database served as the environment for executing SQL queries on the cleaned dataset.

Creating SQL Table

The cleaned dataset was transferred into a database table using Pandas and SQLite integration. This allowed SQL commands to be executed directly on the dataset.

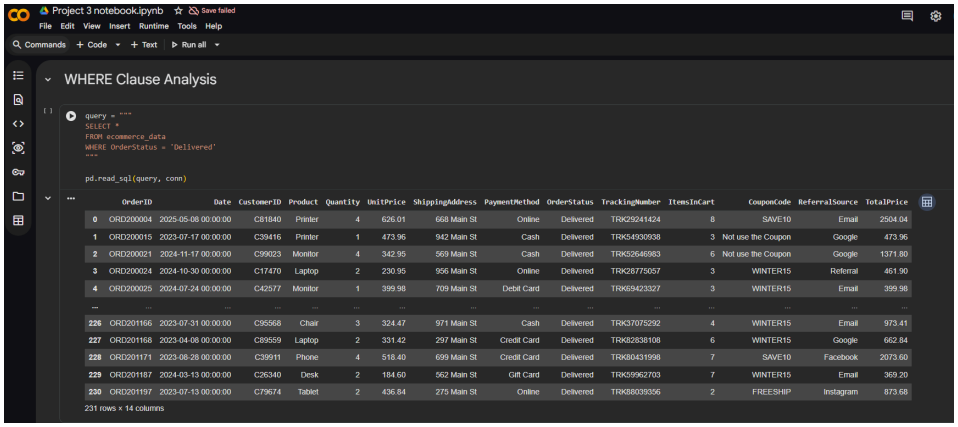
7. SQL Query Analysis

Viewing Dataset Records



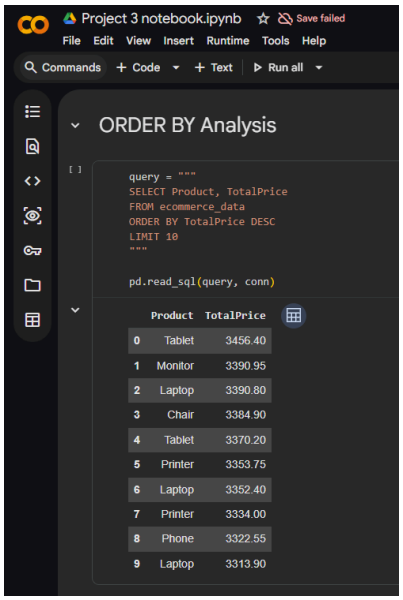
A SELECT query was executed to retrieve sample records from the database table and verify successful data insertion.

Delivered Orders Analysis



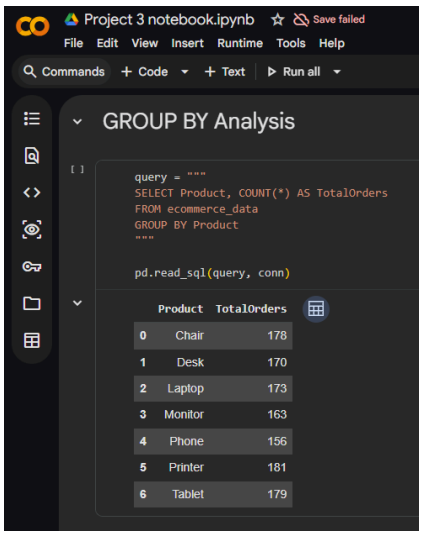
A WHERE clause was used to retrieve only delivered orders. This analysis helped identify successfully completed transactions within the dataset.

Highest Value Products



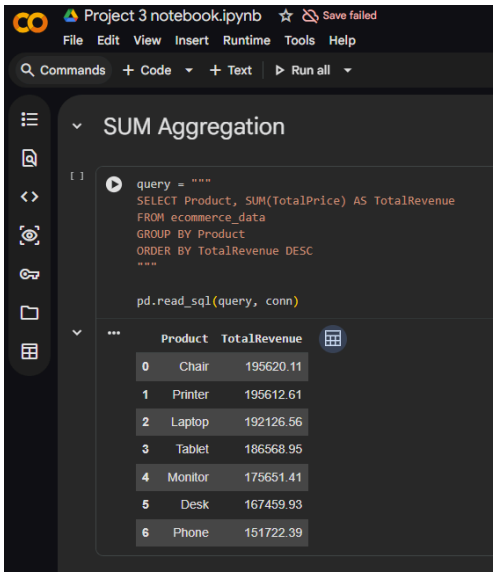
An ORDER BY query was used to sort products based on transaction value. This helped identify products generating the highest sales revenue.

Product Order Analysis



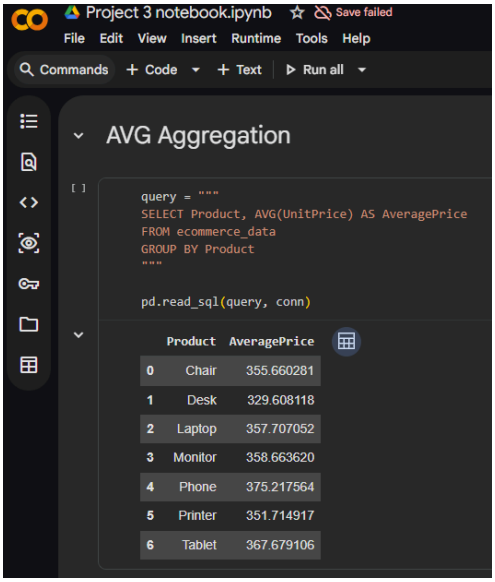
A GROUP BY query was performed to count the number of orders associated with each product. This helped identify the most popular products among customers.

Revenue Analysis



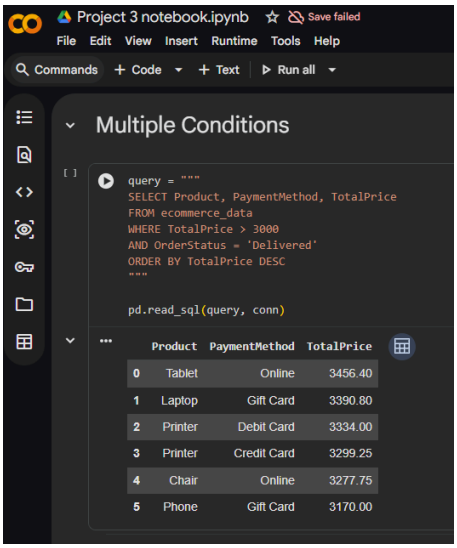
The SUM function was used to calculate total revenue generated by each product. This analysis highlighted the products contributing most to overall sales.

Average Product Price Analysis



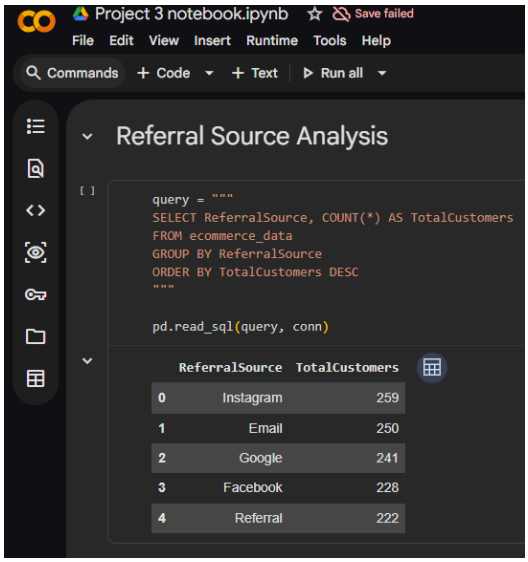
The AVG function was used to calculate the average price of products in the dataset. This provided insights into product pricing patterns.

High-Value Delivered Orders



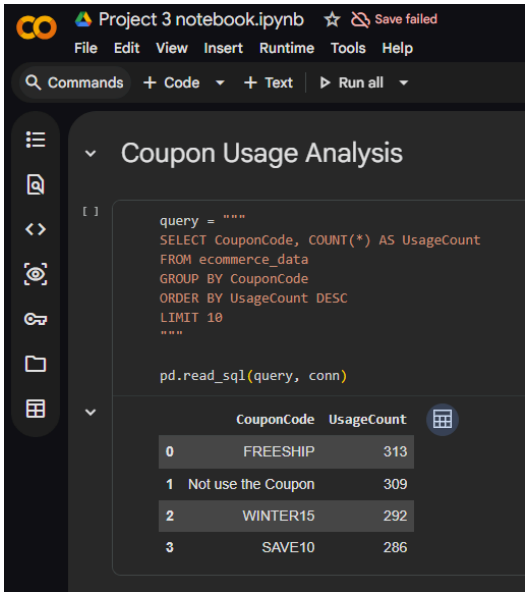
Multiple SQL conditions were used to identify delivered orders with high transaction values. This analysis helped understand premium customer purchases.

Referral Source Analysis



The dataset was analyzed based on referral sources to determine which marketing channels generated the highest customer engagement and transactions.

Coupon Usage Analysis



Coupon usage data was analyzed to evaluate customer participation in promotional campaigns and discount programs.

9. Conclusion

This project helped me learn about the practical application of SQL in data analysis. It was possible to do this by using Python, Pandas, and SQLite to clean, structure, and analyze an e-commerce dataset.

One of the goals of the project was to learn about the theoretical and practical aspects related to databases, SQL queries, and data analysis. The need for SQL knowledge was also evident from this project.